

# Copied Infrared Evidence

Preserve what a bounded Lesson 025 frame can actually prove

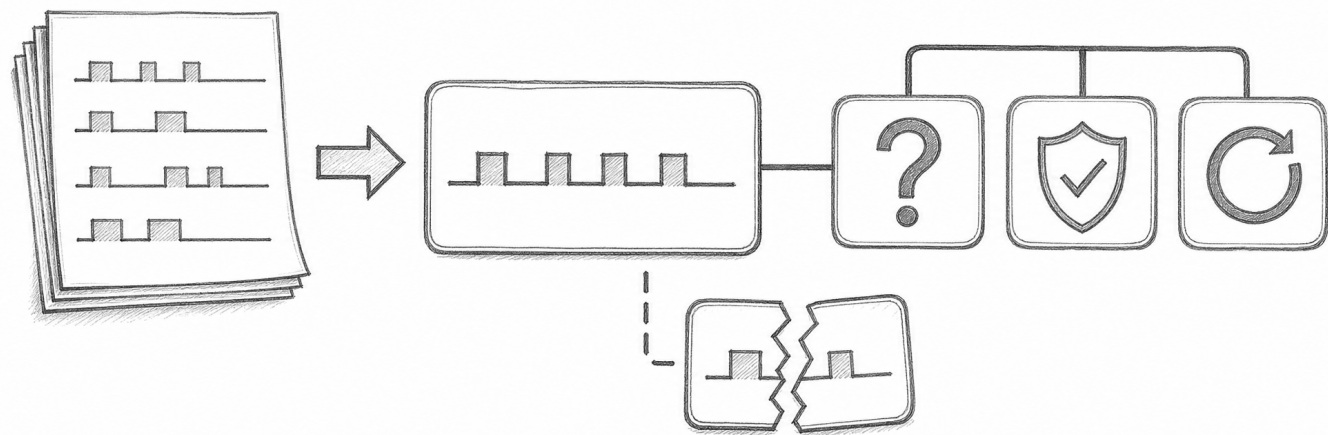
E0 COPIED REPLAY — NO RECEIVER, PIN, OR OPTICAL CLAIM

|              |                                     |          |                          |
|--------------|-------------------------------------|----------|--------------------------|
| Lesson       | 052 — component                     | Time     | 100–140 minutes          |
| Prerequisite | Lesson 025                          | Board    | compile-only Mega replay |
| Evidence     | caller-owned words and result cells | Boundary | no capture hardware      |

**Responsibility.** `CapturedIrEvidence` synchronously copies one borrowed Lesson 025 pulse frame into fixed caller-owned storage, classifies it, and publishes attributable evidence. It does not retain the borrowed view, decode an unknown protocol, operate a receiver, or grant transmission authority.

## 1 Copy before acknowledging the producer

A Lesson 025 view is borrowed. While it is valid, copy every significant duration and mark bit into the configured destination. Reject a frame that is empty, too long, truncated, overflowed, malformed, or not representable in the fixed storage. Only after the copy and classification are complete may the caller acknowledge the producer.



Alternative text: graphite pulse sheets enter one bounded copied record, which branches to question-mark, shield-check, and repeat cards. A broken pulse card below is separated from the accepted path.

The drawing is a conceptual memory-flow mnemonic. It contains no receiver, wiring, carrier, pin assignment, or formal schematic.

## 2 Use categorical evidence, never invented confidence

| Strength          | What the copied record supports                              | What it cannot support                          |
|-------------------|--------------------------------------------------------------|-------------------------------------------------|
| None              | no usable qualified observation                              | protocol or command claim                       |
| ShapeRecognized   | bounded structure resembles the known shape                  | integrity or transmit authority                 |
| IntegrityVerified | the configured decoder verified the known record's integrity | physical origin, user intent, or safe actuation |

Repeat is a distinct disposition, not a stronger command. Unknown, malformed, overflowed, truncated, stale, and repeat records remain inspectable but never inherit the authority of an earlier valid record.

## 3 Bind evidence to its source and generation

Each accepted snapshot retains source kind and ID, configuration revision, qualification epoch, occurrence and observation times, source sequence, decoder disposition, categorical strength, significant word count, and a digest of the complete copied payload. A

same-sequence byte-identical replay is idempotent. Changed content at the same sequence is malformed.

Forward sequence order and elapsed time use unsigned modular half-range comparison. Exact-half-range ambiguity rejects; rollover alone is not a fault. Stale, future, regressing, source-changing, or configuration-changing input does not partly replace the last accepted evidence.

| Case                                   | Predict before replay                                  |
|----------------------------------------|--------------------------------------------------------|
| known valid frame                      | copied view, <b>IntegrityVerified</b> , new generation |
| recognized shape with failed integrity | inspectable, no valid command                          |
| repeat record                          | repeat disposition, no inherited authorization         |
| unknown bounded frame                  | copied for diagnosis, never emission input             |
| overflow or truncation                 | atomic rejection; no partial record                    |
| same sequence, changed pulse           | malformed; accepted history preserved                  |

## 4 Run the deterministic workbook

Use the canonical compile-only Lesson 052 sketch and executable host tests. Predict each named result cell before inspecting it.

1. Initialize with fixed caller storage and a nonzero instance epoch. Predict inactive lifecycle, zero copied words, and no evidence.
2. Admit one known valid synthetic frame. Record disposition, strength, source sequence, evidence generation, and digest.
3. Replay it byte-for-byte at the same sequence, then change one duration. Distinguish idempotence from malformed mutation.
4. Supply shape-recognized, repeat, unknown, malformed, truncated, and overflowed frames. Verify that each remains categorically distinct.
5. Exercise exact freshness boundaries, timestamp rollover, regression, future observation, and half-range ambiguity.
6. Change source ID, configuration revision, and qualification epoch. Verify provenance checks before any snapshot mutation.
7. Reset and restart. Confirm that old evidence and old generations do not silently become current.

| frame | disposition | strength | words | generation |
|-------|-------------|----------|-------|------------|
| _____ | _____       | _____    | _____ | _____      |
| _____ | _____       | _____    | _____ | _____      |
| _____ | _____       | _____    | _____ | _____      |

## 5 Separate acquisition evidence from safe-state evidence

**Acquisition prediction.** Valid extents, non-overlapping caller storage, supported source constraints, and a nonzero epoch allow E0 initialization.

**Acquisition observation.** Host tests inspect lifecycle status, copied-word count, provenance, digest, disposition, strength, and generation in memory.

**Acquisition interpretation.** Success proves bounded copying and classification only. It does not prove a receiver, interrupt, pin, carrier, optical signal, remote identity, or physical Mega execution.

**Safe-state prediction and observation.** Construction, failed initialization, reset, fault, and shutdown publish no current valid evidence and cannot produce emission intent. This is software state, not proof that a future electrical endpoint is unpowered.

## 6 Read the conservative resource evidence

The final canonical Mega 2560 sketch measures 5,530 bytes of flash and 1,096 bytes of static SRAM with Arduino AVR core 1.8.8 and avr-gcc 7.3.0. `sizeof(CapturedIrEvidence)` is 48 bytes. Conservative compiler stack evidence estimates about 197 bytes on the measured deepest analyzed path.

These numbers are compiler evidence for this exact source, toolchain, optimization, and fixture. They are not runtime high-water measurements, proof of stack safety under every interrupt schedule, or a powered-hardware claim. Re-measure whenever the sketch, library, core, compiler, flags, or composition changes.

## 7 Keep E1 explicitly open

An E1 adapter needs an exact receiver specimen, primary evidence, voltage and timing limits, qualified pin and interrupt ownership, a formal authoritative schematic, acquisition rollback, non-Serial observation, and recorded bench acceptance. None is supplied here. Phone-camera observations are not receiver qualification.

## 8 Evidence and next step

Compare the compile-only Mega result cells with the deterministic component tests and the accessible HTML lesson. Record:

- acquisition status: \_\_\_\_\_ evidence generation: \_\_\_\_\_
- copied count: \_\_\_\_\_ disposition: \_\_\_\_\_
- strength: \_\_\_\_\_ safe-state result: \_\_\_\_\_

Lesson 053 accepts only a fixed locally authored symbolic catalog. No `CapturedIrView`, copied pulse, raw duration, unknown command, or repeat record can enter that catalog.